

Jessica E. Snyder, PhD

Work

Universities Space Research Assoc contracted to NASA Ames Research Center

Scientist (2017-2020) *Biological industrial design to outfit human missions to space.*

Worked alongside evolutionary and synthetic biologists to reverse engineer life from extreme environments and then build biotechnology. Project included: Digitizing a sea sponge to 3D print a living filter, a portable device to make protein-based drugs, and field work in Antarctica's South Shetland Islands.

Massachusetts Institute of Technology, Senseable City Lab

Postdoctoral Fellow (2015-2017) *A somewhat anonymized public health census using biology, data, and civic infrastructure.* As part of the Underworlds team, I automated collection of wastewater to measure the bacteria, virus, and chemicals neighborhood by neighborhood. Our geolocated data-set became a heat-map of human health across the city. Instances of the technology were deployed in Boston and Kuwait. The project has become the company BioBot Analytics.

Drexel University, Mechanical Engineering Department

Research Fellow / PhD Candidate (2009-2014) *Invent manufacturing process to build a model of human liver to study drug metabolism.* Engineered a now patented process to 3D print multiple cell types to mimic the anatomical structure of hepatic tissue. With collaborator from NASA Johnson Space Center, tested the metabolism of the 3D printed model of liver in a space-like environment.

Patented Inventions

Heterogeneous filaments, methods of producing the same, scaffolds, methods of producing the same, droplets, and methods of producing the same. Sun, Wei, Qudus Hamid, and Jessica Snyder. Publication Date 02-02-2017
<https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2017019300>

Compositions and Methods for Functionalized Patterning of Tissue

Engineering Substrates Including Bioprinting Cell-Laden Constructs for

Multicompartment Tissue Chambers. Sun, Wei, Jessica Snyder, Robert Chang, Eda Yildirim, Alexander Fridman, and Halim Ayan. U.S. Patent Application 12 872,992, led June 9, 2011. <https://www.freepatentsonline.com/y2011/0136162.html>

Service

Montefiore Health System, New York City

Collaborator (2016 - present) *A clinical trial to measure emotion during painful procedures to test the effectiveness of virtual reality distractions.* The goal is to tailor-make immersive entertainment experiences that can replace opiates and guide patients towards calmer emotional states during hospitalization.

Cell Culture Skills: BSL2 technique, bioprinting, bioreactor (rotary, microfluidic, oscillatory compression/relaxation)

Programming Skills: Digital reconstruction of medical images (ImageJ), data analysis (Python, MatLab), computer aided design and manufacturing CAD/CAM (AutoCAD, Fusion360, SolidWorks, OpenSCAD)

Prototyping Skills: 3D printing, laser cutting, soldering, replica molding (PDMS, gallium), electrospinning, coaxial extrusion, mechanical testing

Education

2009 BS/MS Mech Eng, Drexel University
2014 PhD Mech Eng, Drexel University
2015-17 Postdoc Urban Planning, MIT

Adjunct Faculty

2019 Santa Clara University
2021 Cogswell University of Silicon Valley

Miscellaneous

Blue Marble Space Institute of Science, Affiliate
HISEAS, Analog Astronaut (Selene I)
World Health Summit, New Voices 2016